

CALCULUS AB
SECTION I, Part A
Time—55 minutes
Number of questions—28

A CALCULATOR MAY NOT BE USED ON THIS PART OF THE EXAM.

Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and fill in the corresponding circle on the answer sheet. No credit will be given for anything written in the exam book. Do not spend too much time on any one problem.

1. $\lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x^2 - 4}$ is

2. If $f(x) = x^3 - x^2 + x - 1$, then $f'(2) =$

- 4-

A A

4. Which of the following is an equation of the line tangent to the graph of $x^2 - 3xy = 10$ at the point $(1, -3)$?

(A) $y + 3 = -11(x - 1)$

(B) $y + 3 = -\frac{7}{3}(x - 1)$

(C) $y + 3 = \frac{1}{3}(x - 1)$

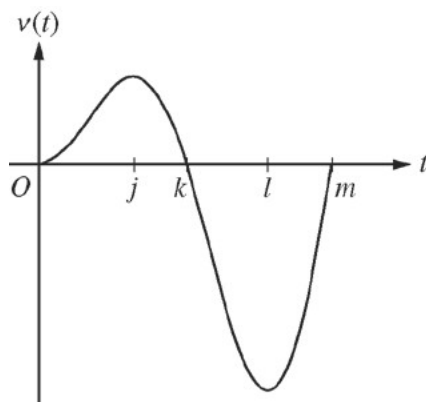
(D) $y + 3 = \frac{7}{3}(x - 1)$

(E) $y + 3 = \frac{11}{3}(x - 1)$

A A

7. Let f be the function given by $f(x) = x^3 - 6x^2 + 8x - 2$. What is the instantaneous rate of change of f at $x = 3$?

- (A) -5 (B) $-\frac{15}{4}$ (C) -1 (D) 6 (E) 17



8. A particle moves along a straight line. The graph of the particle's velocity $v(t)$ at time t is shown above for $0 \leq t \leq m$, where j , k , l , and m are constants. The graph intersects the horizontal axis at $t = 0$, $t = k$, and $t = m$ and has horizontal tangents at $t = j$ and $t = l$. For what values of t is the speed of the particle decreasing?

- (A) $j \leq t \leq l$
 (B) $k \leq t \leq m$
 (C) $j \leq t \leq k$ and $l \leq t \leq m$
 (D) $0 \leq t \leq j$ and $k \leq t \leq l$
 (E) $0 \leq t \leq j$ and $l \leq t \leq m$

A A

11. Let f be the function defined by $f(x) = \int_0^x (2t^3 - 15t^2 + 36t) dt$. On which of the following intervals is the graph of $y = f(x)$ concave down?

(A) $(-\infty, 0)$ only

(B) $(-\infty, 2)$

(C) $(0, \infty)$

(D) $(2, 3)$ only

(E) $(3, \infty)$ only

12. For which of the following does $\lim_{x \rightarrow \infty} f(x) = 0$?

$$\text{I. } f(x) = \frac{\ln x}{x^{99}}$$

II. $f(x) = \frac{e^x}{\ln x}$

III. $f(x) = \frac{x^{99}}{e^x}$

- (A) I only
(B) II only
(C) III only
(D) I and II only
(E) I and III only

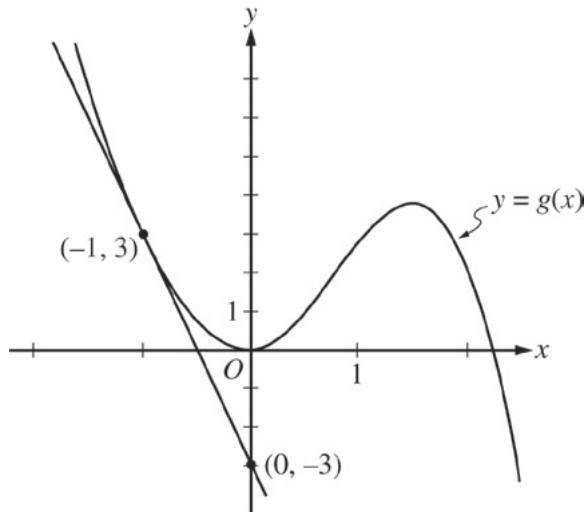
17. If $\lim_{h \rightarrow 0} \frac{\arcsin(a+h) - \arcsin(a)}{h} = 2$, which of the following could be the value of a ?

- (A) $\frac{\sqrt{2}}{2}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\sqrt{3}$ (D) $\frac{1}{2}$ (E) 2

18. If $\ln(2x + y) = x + 1$, then $\frac{dy}{dx} =$

- (A) -2 (B) $2x + y - 2$ (C) $2x + y$ (D) $4x + 2y - 2$ (E) $y - \frac{y}{x}$

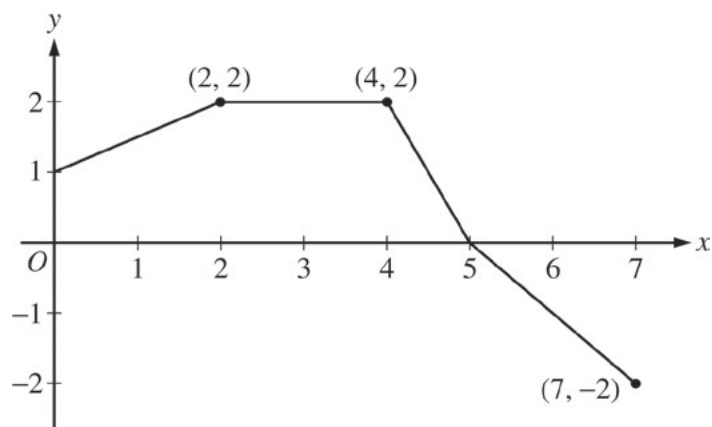
A A



19. The figure above shows the graph of the function g and the line tangent to the graph of g at $x = -1$. Let h be the function given by $h(x) = e^x \cdot g(x)$. What is the value of $h'(-1)$?

- (A) $\frac{9}{e}$ (B) $\frac{-3}{e}$ (C) $\frac{-6}{e}$ (D) $\frac{-6}{e} - \frac{3}{e^2}$ (E) -6

A A



Graph of f

21. The graph of a function f is shown above. What is the value of $\int_0^7 f(x) dx$?
- (A) 6 (B) 8 (C) 10 (D) 14 (E) 18

B**B****B****B****B****B****B****B****B****CALCULUS AB****SECTION I, Part B****Time — 50 minutes****Number of questions — 17**

A GRAPHING CALCULATOR IS REQUIRED FOR SOME QUESTIONS ON
THIS PART OF THE EXAM.

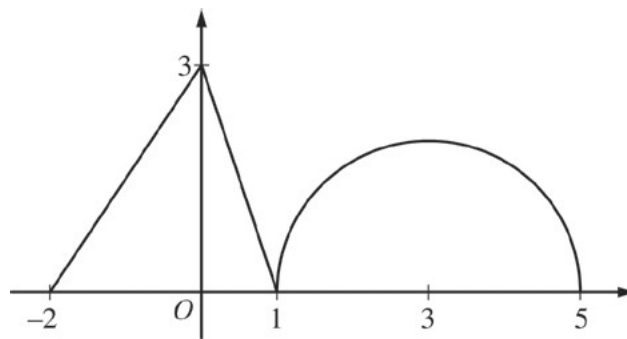
Directions: Solve each of the following problems, using the available space for scratch work. After examining the form of the choices, decide which is the best of the choices given and fill in the corresponding circle on the answer sheet. No credit will be given for anything written in the exam book. Do not spend too much time on any one problem.

BE SURE YOU ARE USING PAGE 3 OF THE ANSWER SHEET TO RECORD YOUR ANSWERS TO QUESTIONS NUMBERED 76–92.

YOU MAY NOT RETURN TO PAGE 2 OF THE ANSWER SHEET.

In this exam:

- (1) The exact numerical value of the correct answer does not always appear among the choices given. When this happens, select from among the choices the number that best approximates the exact numerical value.
- (2) Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.
- (3) The inverse of a trigonometric function f may be indicated using the inverse function notation f^{-1} or with the prefix “arc” (e.g., $\sin^{-1} x = \arcsin x$).

B**B****B****B****B****B****B****B****B**Graph of f

76. The graph of the function f shown above consists of two line segments and a semicircle. Let g be defined by

$$g(x) = \int_0^x f(t) \, dt. \text{ What is the value of } g(5)?$$

- (A) 0 (B) $-1.5 + 2\pi$ (C) 2π (D) $1.5 + 2\pi$ (E) $4.5 + 2\pi$

B**B****B****B****B****B****B****B****B**

77. The volume of a sphere is decreasing at a constant rate of 3 cubic centimeters per second. At the instant when the radius of the sphere is decreasing at a rate of 0.25 centimeter per second, what is the radius of the sphere?

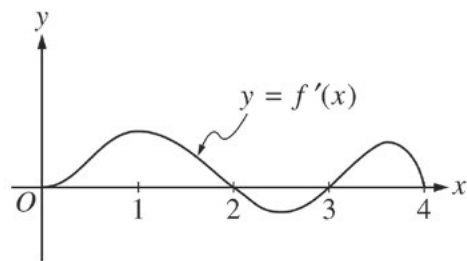
(The volume V of a sphere with radius r is $V = \frac{4}{3}\pi r^3$.)

- (A) 0.141 cm (B) 0.244 cm (C) 0.250 cm (D) 0.489 cm (E) 0.977 cm

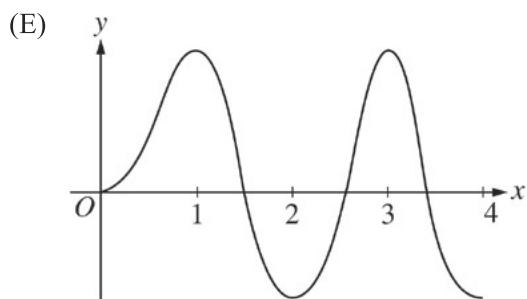
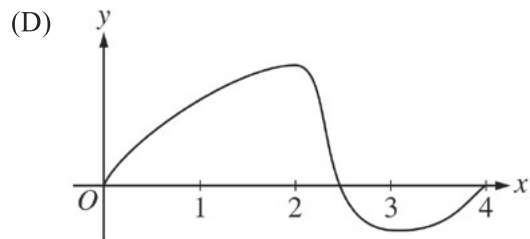
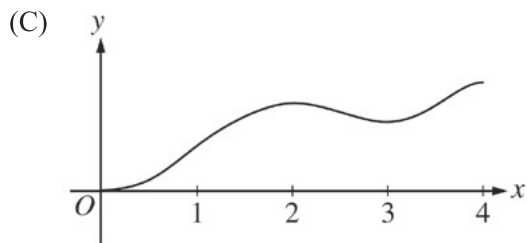
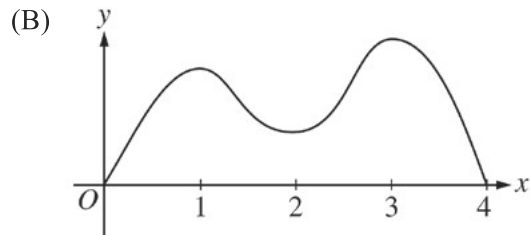
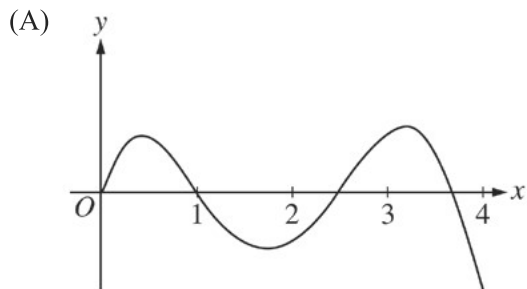
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78. Let f and g be continuous functions such that $\int_0^{10} f(x) \, dx = 21$, $\int_0^{10} \frac{1}{2}g(x) \, dx = 8$, and

$\int_3^{10} (f(x) - g(x)) \, dx = 2$. What is the value of $\int_0^3 (f(x) - g(x)) \, dx$?

- (A) 3 (B) 7 (C) 11 (D) 15 (E) 19

B**B****B****B****B****B****B****B****B**

79. The figure above shows the graph of f' , the derivative of the function f . If $f(0) = 0$, which of the following could be the graph of f ?



B**B****B****B****B****B****B****B****B**

80. For time $t \geq 0$, the position of a particle traveling along a line is given by a differentiable function s . If s is increasing for $0 \leq t < 2$ and s is decreasing for $t > 2$, which of the following is the total distance the particle travels for $0 \leq t \leq 5$?

(A) $s(0) + \int_0^2 s'(t) \, dt - \int_2^5 s'(t) \, dt$

(B) $s(0) + \int_2^5 s'(t) \, dt - \int_0^2 s'(t) \, dt$

(C) $\int_2^5 s'(t) \, dt - \int_0^2 s'(t) \, dt$

(D) $\left| \int_0^5 s'(t) \, dt \right|$

(E) $\int_0^5 |s'(t)| \, dt$

B**B****B****B****B****B****B****B****B**

81. A cup of tea is cooling in a room that has a constant temperature of 70 degrees Fahrenheit ($^{\circ}\text{F}$). If the initial temperature of the tea, at time $t = 0$ minutes, is 200°F and the temperature of the tea changes at the rate $R(t) = -6.89e^{-0.053t}$ degrees Fahrenheit per minute, what is the temperature, to the nearest degree, of the tea after 4 minutes?

(A) 175°F (B) 130°F (C) 95°F (D) 70°F (E) 45°F

B**B****B****B****B****B****B****B****B**

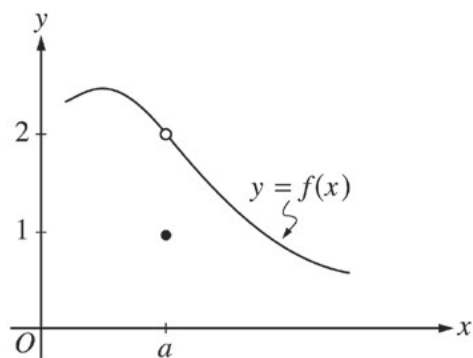
82. The derivative of the function f is given by $f'(x) = x^3 - 4\sin(x^2) + 1$. On the interval $(-2.5, 2.5)$, at which of the following values of x does f have a relative maximum?

- (A) -1.970 and 0
(B) -1.467 and 1.075
(C) -0.475 , 0.542 , and 1.396
(D) -0.475 and 1.396 only
(E) 0.542 only

x	0	0.5	1	1.5	2	2.5	3
$f(x)$	0	4	10	18	28	40	54

83. The table above gives selected values for a continuous function f . If f is increasing over the closed interval $[0, 3]$, which of the following could be the value of $\int_0^3 f(x) dx$?

- (A) 50 (B) 62 (C) 77 (D) 100 (E) 154

B**B****B****B****B****B****B****B****B**

84. The graph of a function f is shown in the figure above. Which of the following statements is true?

- (A) $f(a) = 2$
- (B) f is continuous at $x = a$.
- (C) $\lim_{x \rightarrow a} f(x) = 1$
- (D) $\lim_{x \rightarrow a} f(x) = 2$
- (E) $\lim_{x \rightarrow a} f(x)$ does not exist.

85. A particle moves along the x -axis so that at time $t \geq 0$ its position is given by $x(t) = \cos \sqrt{t}$. What is the velocity of the particle at the first instance the particle is at the origin?

- (A) -1
- (B) -0.624
- (C) -0.318
- (D) 0
- (E) 0.065

B**B****B****B****B****B****B****B****B**

86. If $f'(x) > 0$ for all x and $f''(x) < 0$ for all x , which of the following could be a table of values for f ?

(A)

x	$f(x)$
-1	4
0	3
1	1

(B)

x	$f(x)$
-1	4
0	4
1	4

(C)

x	$f(x)$
-1	4
0	5
1	6

(D)

x	$f(x)$
-1	4
0	5
1	7

(E)

x	$f(x)$
-1	4
0	6
1	7

B**B****B****B****B****B****B****B****B**

87. Let f be the function with first derivative given by $f'(x) = (3 - 2x - x^2)\sin(2x - 3)$. How many relative extrema does f have on the open interval $-4 < x < 2$?

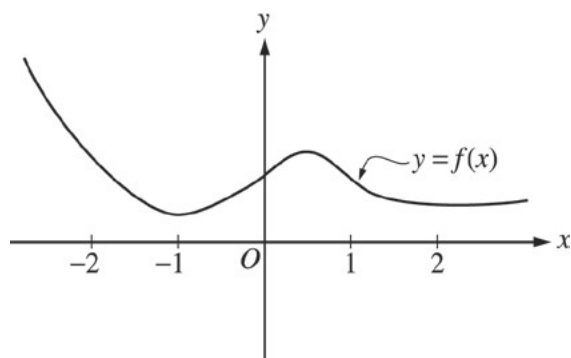
(A) Two

(B) Three

(C) Four

(D) Five

(E) Six

B**B****B****B****B****B****B****B****B**

88. The graph of a twice-differentiable function f is shown in the figure above. Which of the following is true?

- (A) $f'(-1) < f'(1) < f'(0)$
- (B) $f'(-1) < f'(0) < f'(1)$
- (C) $f'(0) < f'(-1) < f'(1)$
- (D) $f'(1) < f'(-1) < f'(0)$
- (E) $f'(1) < f'(0) < f'(-1)$

B**B****B****B****B****B****B****B****B**

89. What is the volume of the solid generated when the region bounded by the graph of $x = \sqrt{y - 2}$ and the lines $x = 0$ and $y = 5$ is revolved about the y -axis?

- (A) 3.464 (B) 4.500 (C) 7.854 (D) 10.883 (E) 14.137

90. The population P of a city grows according to the differential equation $\frac{dP}{dt} = kP$, where k is a constant and t is measured in years. If the population of the city doubles every 12 years, what is the value of k ?

- (A) 0.058 (B) 0.061 (C) 0.167 (D) 0.693 (E) 8.318

B**B****B****B****B****B****B****B****B**

91. The function f is continuous and $\int_0^8 f(u) \, du = 6$. What is the value of $\int_1^3 xf(x^2 - 1) \, dx$?

- (A) $\frac{3}{2}$ (B) 3 (C) 6 (D) 12 (E) 24

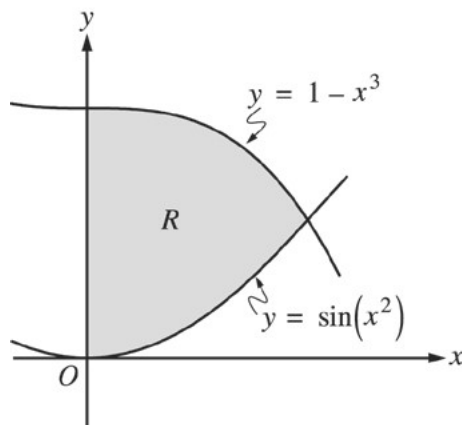
92. The function f is defined for all x in the closed interval $[a, b]$. If f does not attain a maximum value on $[a, b]$, which of the following must be true?

- (A) f is not continuous on $[a, b]$.
(B) f is not bounded on $[a, b]$.
(C) f does not attain a minimum value on $[a, b]$.
(D) The graph of f has a vertical asymptote in the interval $[a, b]$.
(E) The equation $f'(x) = 0$ does not have a solution in the interval $[a, b]$.

CALCULUS AB
SECTION II, Part A
Time—30 minutes
Number of problems—2

A graphing calculator is required for these problems.

GO ON TO THE NEXT PAGE.



1. Let R be the shaded region in the first quadrant enclosed by the y -axis and the graphs of $y = 1 - x^3$ and $y = \sin(x^2)$, as shown in the figure above.
 - (a) Find the area of R .

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1**1****1****1****1****1****1****1****1****1**

- (b) A horizontal line, $y = k$, is drawn through the point where the graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect. Find k and determine whether this line divides R into two regions of equal area. Show the work that leads to your conclusion.

-
- (c) Find the volume of the solid generated when R is revolved about the line $y = -3$.

2. The penguin population on an island is modeled by a differentiable function P of time t , where $P(t)$ is the number of penguins and t is measured in years, for $0 \leq t \leq 40$. There are 100,000 penguins on the island at time $t = 0$. The birth rate for the penguins on the island is modeled by

$$B(t) = 1000e^{0.06t} \text{ penguins per year}$$

and the death rate for the penguins on the island is modeled by

$$D(t) = 250e^{0.1t} \text{ penguins per year.}$$

- (a) What is the rate of change of the penguin population on the island at time $t = 0$?

-
- (b) To the nearest whole number, what is the penguin population on the island at time $t = 40$?

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2**2****2****2****2****2****2****2****2****2**

- (c) To the nearest whole number, what is the average rate of change of the penguin population on the island for $0 \leq t \leq 40$?

-
- (d) To the nearest whole number, find the absolute minimum penguin population and the absolute maximum penguin population on the island for $0 \leq t \leq 40$. Show the analysis that leads to your answers.

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CALCULUS AB
SECTION II, Part B
Time—60 minutes
Number of problems—4

No calculator is allowed for these problems.

DO NOT BREAK THE SEALS UNTIL YOU ARE TOLD TO DO SO.

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NO CALCULATOR ALLOWED

t (days)	0	10	22	30
$W'(t)$ (GL per day)	0.6	0.7	1.0	0.5

3. The twice-differentiable function W models the volume of water in a reservoir at time t , where $W(t)$ is measured in gigaleters (GL) and t is measured in days. The table above gives values of $W'(t)$ sampled at various times during the time interval $0 \leq t \leq 30$ days. At time $t = 30$, the reservoir contains 125 gigaleters of water.
- (a) Use the tangent line approximation to W at time $t = 30$ to predict the volume of water $W(t)$, in gigaleters, in the reservoir at time $t = 32$. Show the computations that lead to your answer.

- (b) Use a left Riemann sum, with the three subintervals indicated by the data in the table, to approximate $\int_0^{30} W'(t) dt$. Use this approximation to estimate the volume of water $W(t)$, in gigaleters, in the reservoir at time $t = 0$. Show the computations that lead to your answer.

3**3****3****3****3****3****3****3****3****3****NO CALCULATOR ALLOWED**

(c) Explain why there must be at least one time t , other than $t = 10$, such that $W'(t) = 0.7$ GL/day.

(d) The equation $A = 0.3W^{2/3}$ gives the relationship between the area A , in square kilometers, of the surface of the reservoir, and the volume of water $W(t)$, in giga-liters, in the reservoir. Find the instantaneous rate of change of A , in square kilometers per day, with respect to t when $t = 30$ days.

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NO CALCULATOR ALLOWED

4. Let f be the function given by $f(x) = (x^2 - 2x - 1)e^x$.

(a) Find $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$.

$$\lim_{x \rightarrow \infty} f(x) = \underline{\hspace{2cm}}$$

$$\lim_{x \rightarrow -\infty} f(x) = \underline{\hspace{2cm}}$$

(b) Find the intervals on which f is increasing. Show the analysis that leads to your answer.

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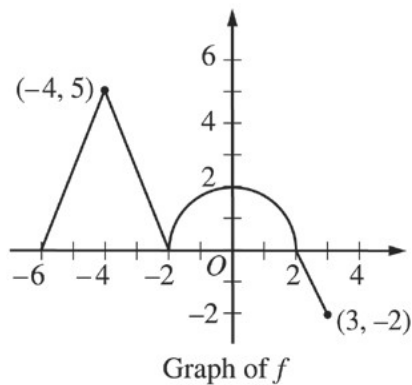
NO CALCULATOR ALLOWED

- (c) Find the intervals on which the graph of f is concave down. Show the analysis that leads to your answer.

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NO CALCULATOR ALLOWED



5. The graph of the continuous function f , consisting of three line segments and a semicircle, is shown above.

Let g be the function given by $g(x) = \int_{-2}^x f(t) dt$.

(a) Find $g(-6)$ and $g(3)$.

(b) Find $g'(0)$.

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NO CALCULATOR ALLOWED

- (c) Find all values of x on the open interval $-6 < x < 3$ for which the graph of g has a horizontal tangent. Determine whether g has a local maximum, a local minimum, or neither at each of these values. Justify your answers.

-
- (d) Find all values of x on the open interval $-6 < x < 3$ for which the graph of g has a point of inflection. Explain your reasoning.

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6**6****6****6****6****6****6****6****6****6****NO CALCULATOR ALLOWED**

6. Let f be a function with $f(2) = -8$ such that for all points (x, y) on the graph of f , the slope is given by $\frac{3x^2}{y}$.
- (a) Write an equation of the line tangent to the graph of f at the point where $x = 2$ and use it to approximate $f(1.8)$.

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NO CALCULATOR ALLOWED

- (b) Find an expression for $y = f(x)$ by solving the differential equation $\frac{dy}{dx} = \frac{3x^2}{y}$ with the initial condition $f(2) = -8$.

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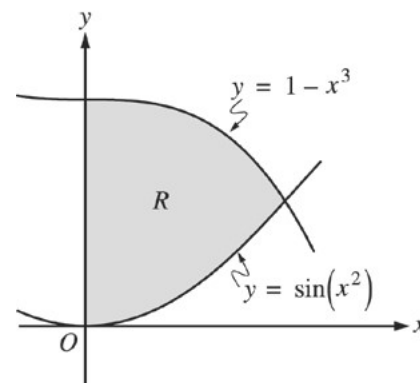
Answer Key for AP Calculus AB Practice Exam, Section I

Question 1: D	Question 24: A
Question 2: B	Question 25: C
Question 3: B	Question 26: E
Question 4: E	Question 27: D
Question 5: D	Question 28: B
Question 6: B	Question 76: D
Question 7: C	Question 77: E
Question 8: C	Question 78: A
Question 9: D	Question 79: C
Question 10: A	Question 80: E
Question 11: D	Question 81: A
Question 12: E	Question 82: E
Question 13: E	Question 83: B
Question 14: A	Question 84: D
Question 15: C	Question 85: C
Question 16: D	Question 86: E
Question 17: B	Question 87: E
Question 18: B	Question 88: D
Question 19: B	Question 89: E
Question 20: D	Question 90: A
Question 21: A	Question 91: B
Question 22: E	Question 92: A
Question 23: B	

AP[®] CALCULUS AB
2013 SCORING GUIDELINES

Question 1

Let R be the shaded region in the first quadrant enclosed by the y -axis and the graphs of $y = 1 - x^3$ and $y = \sin(x^2)$, as shown in the figure above.



- (a) Find the area of R .
- (b) A horizontal line, $y = k$, is drawn through the point where the graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect. Find k and determine whether this line divides R into two regions of equal area. Show the work that leads to your conclusion.
- (c) Find the volume of the solid generated when R is revolved about the line $y = -3$.

The graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect in the first quadrant at the point $(A, B) = (0.764972, 0.552352)$.

(a)
$$\text{Area} = \int_0^A (1 - x^3 - \sin(x^2)) dx$$

$$= 0.533 \text{ (or } 0.534\text{)}$$

(b) $k = B = 0.552352$

$$\int_0^A (1 - x^3 - k) dx = 0.257 \text{ (or } 0.256\text{)}$$

$$\int_0^A (k - \sin(x^2)) dx = 0.277 \text{ (or } 0.276\text{)}$$

The two regions have unequal areas.

(c)
$$\text{Volume} = \pi \int_0^A \left((1 - x^3 + 3)^2 - (\sin(x^2) + 3)^2 \right) dx$$

$$= 11.841 \text{ (or } 11.840\text{)}$$

1 : correct limits in an integral in (a), (b), or (c)

2 : $\begin{cases} 1 : \text{integrand} \\ 1 : \text{answer} \end{cases}$

3 : $\begin{cases} 1 : \text{integral(s) with } k \text{ value} \\ 1 : \text{value(s) of integral(s)} \\ 1 : \text{conclusion tied to part (a)} \\ \text{or comparison of two integrals} \end{cases}$

Note: Stating k value only does not earn a point.

3 : $\begin{cases} 2 : \text{integrand} \\ 1 : \text{answer} \end{cases}$

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Question 2

The penguin population on an island is modeled by a differentiable function P of time t , where $P(t)$ is the number of penguins and t is measured in years, for $0 \leq t \leq 40$. There are 100,000 penguins on the island at time $t = 0$. The birth rate for the penguins on the island is modeled by

$$B(t) = 1000e^{0.06t} \text{ penguins per year}$$

and the death rate for the penguins on the island is modeled by

$$D(t) = 250e^{0.1t} \text{ penguins per year.}$$

- What is the rate of change of the penguin population on the island at time $t = 0$?
- To the nearest whole number, what is the penguin population on the island at time $t = 40$?
- To the nearest whole number, what is the average rate of change of the penguin population on the island for $0 \leq t \leq 40$?
- To the nearest whole number, find the absolute minimum penguin population and the absolute maximum penguin population on the island for $0 \leq t \leq 40$. Show the analysis that leads to your answers.

(a) $P'(0) = B(0) - D(0) = 1000 - 250 = 750$ penguins per year

1 : answer

(b)
$$P(40) = 100000 + \int_0^{40} (B(t) - D(t)) dt$$

$$= 100000 + 33057.56459$$

There are 133,058 penguins on the island.

3 : $\begin{cases} 1 : \text{limits} \\ 1 : \text{integrand} \\ 1 : \text{answer} \end{cases}$

(c)
$$\frac{1}{40} \int_0^{40} (B(t) - D(t)) dt = 826.439$$

OR

$$\frac{P(40) - P(0)}{40 - 0} = \frac{133058 - 100000}{40} = 826.45$$

The average rate of change is 826 penguins per year.

1 : answer

(d) $B(t) - D(t) = 0$

$$1000e^{0.06t} = 250e^{0.1t} \Rightarrow t = A = \frac{\ln 4}{0.04} = 34.657359$$

The absolute minimum and absolute maximum occur at a critical point or at an endpoint.

$$P(0) = 100000$$

$$P(A) = 100000 + \int_0^A (B(t) - D(t)) dt = 139166.667$$

$$P(40) = 133058$$

The minimum population is 100,000 and the maximum population is 139,167 penguins.

4 : $\begin{cases} 1 : B(t) - D(t) = 0 \\ 1 : \text{solves for } t \\ 1 : \text{minimum value} \\ 1 : \text{maximum value} \end{cases}$

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Question 3

t (days)	0	10	22	30
$W'(t)$ (GL per day)	0.6	0.7	1.0	0.5

The twice-differentiable function W models the volume of water in a reservoir at time t , where $W(t)$ is measured in gigaliters (GL) and t is measured in days. The table above gives values of $W'(t)$ sampled at various times during the time interval $0 \leq t \leq 30$ days. At time $t = 30$, the reservoir contains 125 gigaliters of water.

- (a) Use the tangent line approximation to W at time $t = 30$ to predict the volume of water $W(t)$, in gigaliters, in the reservoir at time $t = 32$. Show the computations that lead to your answer.
- (b) Use a left Riemann sum, with the three subintervals indicated by the data in the table, to approximate $\int_0^{30} W'(t) dt$. Use this approximation to estimate the volume of water $W(t)$, in gigaliters, in the reservoir at time $t = 0$. Show the computations that lead to your answer.
- (c) Explain why there must be at least one time t , other than $t = 10$, such that $W'(t) = 0.7$ GL/day.
- (d) The equation $A = 0.3W^{2/3}$ gives the relationship between the area A , in square kilometers, of the surface of the reservoir, and the volume of water $W(t)$, in gigaliters, in the reservoir. Find the instantaneous rate of change of A , in square kilometers per day, with respect to t when $t = 30$ days.

- (a) An equation of the tangent line is $y = 0.5(t - 30) + 125$.
 $W(32) \approx 0.5(32 - 30) + 125 = 126$

1 : answer

- (b) $\int_0^{30} W'(t) dt \approx (10)(0.6) + (12)(0.7) + (8)(1.0) = 22.4$
 $W(0) = W(30) - \int_0^{30} W'(t) dt = 125 - 22.4 = 102.6$

3 : $\begin{cases} 1 : \text{left Riemann sum} \\ 1 : \text{approximation} \\ 1 : \text{answer} \end{cases}$

- (c) W' is differentiable $\Rightarrow W'$ is continuous.

$$W'(30) = 0.5 < 0.7 < 1.0 = W'(22)$$

By the Intermediate Value Theorem, there must be at least one time t , $22 \leq t \leq 30$, such that $W'(t) = 0.7$.

2 : explanation

- (d) $\frac{dA}{dt} = (0.3)\frac{2}{3}W^{-1/3} \cdot \frac{dW}{dt} = \frac{0.2}{\sqrt[3]{W}} \cdot \frac{dW}{dt}$

$$\left. \frac{dA}{dt} \right|_{t=30} = \frac{0.2}{\sqrt[3]{125}} \cdot 0.5 = 0.02$$

3 : $\begin{cases} 2 : \frac{dA}{dt} \\ 1 : \text{answer} \end{cases}$

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Question 4

Let f be the function given by $f(x) = (x^2 - 2x - 1)e^x$.

- (a) Find $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$.
- (b) Find the intervals on which f is increasing. Show the analysis that leads to your answer.
- (c) Find the intervals on which the graph of f is concave down. Show the analysis that leads to your answer.

(a) $\lim_{x \rightarrow \infty} f(x) = \infty$ or does not exist

$$\lim_{x \rightarrow -\infty} f(x) = 0$$

(b) $f'(x) = (2x - 2)e^x + (x^2 - 2x - 1)e^x$
 $= (x^2 - 3)e^x$

$$f'(x) = 0 \text{ when } x = -\sqrt{3}, x = \sqrt{3}$$

$$f'(x) > 0 \text{ for } -\infty < x < -\sqrt{3} \text{ and } \sqrt{3} < x < \infty.$$

$$f \text{ is increasing on the intervals } -\infty < x \leq -\sqrt{3} \text{ and } \sqrt{3} \leq x < \infty.$$

(c) $f''(x) = 2xe^x + (x^2 - 3)e^x$
 $= (x^2 + 2x - 3)e^x = (x + 3)(x - 1)e^x$

$$f''(x) < 0 \text{ for } -3 < x < 1$$

The graph of f is concave down on the interval $-3 < x < 1$.

1 : answers

4 : $\begin{cases} 2 : f'(x) \\ 1 : \text{analysis} \\ 1 : \text{intervals} \end{cases}$

4 : $\begin{cases} 2 : f''(x) \\ 1 : \text{analysis} \\ 1 : \text{interval} \end{cases}$

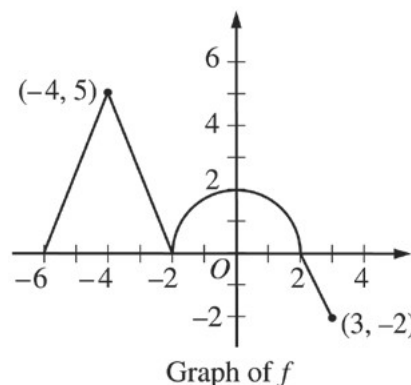
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Question 5

The graph of the continuous function f , consisting of three line segments and a semicircle, is shown above. Let g be the function given by

$$g(x) = \int_{-2}^x f(t) \, dt.$$

- (a) Find $g(-6)$ and $g(3)$.
- (b) Find $g'(0)$.
- (c) Find all values of x on the open interval $-6 < x < 3$ for which the graph of g has a horizontal tangent. Determine whether g has a local maximum, a local minimum, or neither at each of these values. Justify your answers.
- (d) Find all values of x on the open interval $-6 < x < 3$ for which the graph of g has a point of inflection. Explain your reasoning.



(a) $g(-6) = \int_{-2}^{-6} f(t) \, dt = -\int_{-6}^{-2} f(t) \, dt = -\frac{1}{2} \cdot 4 \cdot 5 = -10$

$$g(3) = \int_{-2}^3 f(t) \, dt = \frac{1}{2} \pi \cdot 2^2 - \frac{1}{2} \cdot 1 \cdot 2 = 2\pi - 1$$

(b) $g'(0) = f(0) = 2$

- (c) The graph of g has a horizontal tangent at $x = -2$ and $x = 2$ where $g'(x) = f(x) = 0$.

The graph of g has neither a local maximum nor a local minimum at $x = -2$ because $g'(x) = f(x)$ does not change sign at $x = -2$.

The graph of g has a local maximum at $x = 2$ because $g'(x) = f(x)$ changes sign from positive to negative at $x = 2$.

- (d) The graph of g has a point of inflection at $x = -4$, $x = -2$, and $x = 0$.

$g'(x) = f(x)$ changes from increasing to decreasing at $x = -4$ and $x = 0$, and changes from decreasing to increasing at $x = -2$.

OR

$g''(x) = f'(x)$ changes from positive to negative at $x = -4$ and $x = 0$, and changes from negative to positive at $x = -2$.

$$2 : \begin{cases} 1 : g(-6) \\ 1 : g(3) \end{cases}$$

$$1 : g'(0)$$

$$3 : \begin{cases} 1 : \text{horizontal tangent at } x = -2 \\ \text{and } x = 2 \\ 2 : \text{answers with justifications} \end{cases}$$

$$3 : \begin{cases} 2 : \text{values of } x \\ 1 : \text{explanation} \end{cases}$$

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Question 6

Let f be a function with $f(2) = -8$ such that for all points (x, y) on the graph of f , the slope is given by $\frac{3x^2}{y}$.

- (a) Write an equation of the line tangent to the graph of f at the point where $x = 2$ and use it to approximate $f(1.8)$.
- (b) Find an expression for $y = f(x)$ by solving the differential equation $\frac{dy}{dx} = \frac{3x^2}{y}$ with the initial condition $f(2) = -8$.

(a) Slope = $\frac{(3)(4)}{-8} = -\frac{3}{2}$

An equation for the tangent line is $y = -\frac{3}{2}(x - 2) - 8$.

$$f(1.8) \approx -\frac{3}{2}(1.8 - 2) - 8 = -7.7$$

(b) $\int y \, dy = \int 3x^2 \, dx$

$$\frac{1}{2}y^2 = x^3 + C$$

$$\frac{1}{2}(-8)^2 = 2^3 + C \Rightarrow C = 24$$

$$y^2 = 2(x^3 + 24) = 2x^3 + 48$$

$$y = -\sqrt{2x^3 + 48}$$

Note: This solution is valid for $x > -\sqrt[3]{24}$.

3 : $\left\{ \begin{array}{l} 1 : \text{slope} \\ 1 : \text{tangent line equation} \\ 1 : \text{approximation} \end{array} \right.$

6 : $\left\{ \begin{array}{l} 1 : \text{separation of variables} \\ 2 : \text{antiderivatives} \\ 1 : \text{constant of integration} \\ 1 : \text{uses initial condition} \\ 1 : \text{solves for } y \end{array} \right.$

Note: max 3/6 [1-2-0-0-0] if no constant of integration

Note: 0/6 if no separation of variables